

RENSJON, Sweden

WMO: 020310

Lat: 68.073N

Lon: 19.835E

Elev: 493

StdP: 95.54

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.4	-28.7	-34.7	0.1	-31.4	-31.7	0.2	-28.6	13.9	-4.7	12.4	-5.2	0.3	190	0.790

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	22.7	14.7	20.4	13.9	18.3	12.9	16.1	20.8	14.9	18.9	13.8	17.0	3.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.2	10.7	17.2	13.1	10.0	16.4	12.0	9.3	15.5	46.7	21.0	43.3	19.0	40.2	17.0	19.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.4	8.0	-35.9	25.0	2.1	2.0	-37.4	26.4	-38.6	27.6	-39.8	28.7	-41.3	30.1		
(5)			-35.5	17.5	2.5	1.2	-37.3	18.3	-38.7	19.0	-40.1	19.7	-41.9	20.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.4	-13.3	-13.1	-8.9	-3.1	3.6	8.9	12.6	10.5	5.6	-1.7	-8.1	-10.1		
	DBStd	10.48	7.69	7.74	5.96	4.48	4.00	3.64	3.23	3.11	3.31	4.76	6.02	7.25		
	HDD10.0	4324	723	648	587	394	202	64	12	32	134	363	542	623		
	HDD18.3	7192	981	881	846	644	456	284	181	243	381	621	792	882		
	CDD10.0	177	0	0	0	0	5	31	91	47	4	0	0	0		
	CDD18.3	2	0	0	0	0	0	0	2	1	0	0	0	0		
	CDH23.3	40	0	0	0	0	2	7	28	4	0	0	0	0		
	CDH26.7	3	0	0	0	0	0	1	3	0	0	0	0	0		
(14) Wind	WSAvg	3.1	2.8	2.9	3.5	3.1	3.4	3.5	3.1	2.8	3.2	2.9	2.6	2.9		
(15) Precipitation	PrecAvg	458	23	23	19	23	36	47	80	64	42	43	31	26		
	PrecMax	639	52	39	43	48	95	99	225	134	103	114	58	63		
	PrecMin	287	3	3	8	6	0	0	32	16	13	17	8	9		
	PrecStd	103	12	12	9	13	25	30	47	31	25	24	16	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	5.3	4.8	10.5	22.0	23.9	26.4	23.5	17.1	10.6	6.7	5.5		
		MCWB	1.5	2.4	1.7	5.6	11.7	14.0	16.8	15.9	12.2	7.5	4.0	2.9		
	2%	DB	1.7	2.4	2.8	6.8	17.1	20.1	23.6	20.9	14.3	8.4	3.5	3.0		
		MCWB	0.0	0.3	0.6	3.5	9.7	12.1	15.6	14.9	10.5	6.2	2.0	0.9		
	5%	DB	-0.7	0.2	1.3	5.1	13.4	17.7	21.3	18.5	12.5	6.5	1.4	0.8		
		MCWB	-2.0	-1.4	-0.5	2.3	8.2	11.4	14.8	13.9	9.6	4.9	0.3	-0.7		
	10%	DB	-2.9	-2.0	-0.3	3.6	10.3	15.6	19.1	16.4	11.2	4.6	-0.2	-0.8		
		MCWB	-3.8	-3.2	-2.0	1.3	6.2	10.5	13.8	12.7	8.9	3.4	-1.1	-2.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	2.7	2.1	5.7	12.2	15.5	18.0	17.7	12.7	7.9	4.2	3.1		
		MCDB	3.1	5.1	4.4	10.0	20.0	21.0	23.1	22.2	15.3	9.9	6.3	5.2		
	2%	WB	0.0	0.3	0.8	3.8	10.3	13.7	16.6	15.5	11.2	6.3	2.1	1.1		
		MCDB	1.6	2.3	2.5	6.5	17.0	17.4	22.0	19.3	13.3	8.2	3.3	2.7		
	5%	WB	-2.0	-1.3	-0.5	2.5	8.3	12.4	15.5	14.4	10.2	5.0	0.4	-0.7		
		MCDB	-0.9	0.2	1.2	4.7	13.0	16.9	20.1	17.9	11.8	6.5	1.3	0.6		
	10%	WB	-3.9	-3.2	-1.9	1.4	6.5	11.0	14.5	13.1	9.2	3.5	-0.9	-2.1		
		MCDB	-2.8	-2.1	-0.4	3.4	10.1	15.0	18.1	15.9	10.8	4.5	-0.3	-1.1		
(35) Mean Daily Temperature Range	MDBR		9.2	9.7	10.0	9.1	7.8	8.1	8.8	8.7	7.4	6.7	8.0	8.7		
	5% DB	MCDBR	9.0	9.1	9.3	9.4	13.9	12.2	12.9	12.7	10.2	8.1	8.1	8.3		
		MCWBR	7.9	7.6	7.8	6.5	7.5	5.8	6.0	7.2	6.7	6.1	6.9	7.0		
	5% WB	MCDBR	8.9	8.9	9.1	8.6	13.3	11.0	11.1	11.5	8.2	7.7	7.6	8.3		
		MCWBR	8.0	7.6	7.7	5.9	7.4	5.8	5.7	6.9	5.8	5.9	6.7	7.2		
(40) Clear-Sky Solar Irradiance	taub		0.165	0.225	0.265	0.292	0.311	0.316	0.323	0.310	0.280	0.243	0.170	N/A		
	taud		1.932	2.144	2.235	2.276	2.435	2.519	2.526	2.539	2.584	2.323	1.977	N/A		
	Ebn at Noon		136	601	783	862	886	891	871	845	792	581	116	0		
	Edh at Noon		8	44	74	95	92	88	85	75	55	36	7	0		
(44) All-Sky Solar Radiation	RadAvg		0.04	0.46	1.89	3.84	4.97	4.96	4.58	3.32	1.77	0.58	0.09	0.00		
	RadStd		0.00	0.05	0.11	0.18	0.39	0.40	0.44	0.29	0.18	0.08	0.01	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page